

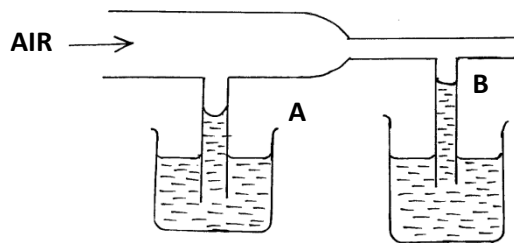
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PHYSICS PAPER 1

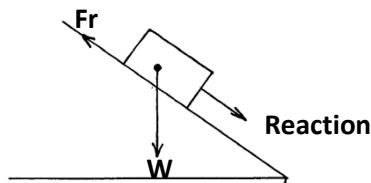
MARKING SCHEME

1. Volume of metal = volume of water displaced
 $= 10.5 \times 3.5$
 $= 36.75 \text{ cm}^3$
 Density = mass / volume
 $= 250 / 36.75$
 $= 6.802 \text{ g/cm}^3$

2 i)



- ii) The pressure at the end of the pipe B is lower than that of A hence water rise more in B than A the air moves at a higher velocity reducing the pressure at B while it moves at a lower /reduced velocity increasing pressure at A
- 3) This is to increase time for collision. This reduces rate of change of momentum: hence impulsive force is reduced
- 4) Density of air decreases making up thrust to be equal to weight of balloon
- 5) Wax on dull surface melts first
 Dull surface is a good absorber of radiant heat than polished absorber
- 6) pressure due to Mercury column = pressure difference to air column
 $\rho_m h_m = \rho_a h_a$;
 $13600 \times 15 = 130 h_a$;
 $H_a = 13600 \times 15 / 130$
 $= 1569.2 \text{ m}$;



- ii) $F_r \rightarrow$ increase
 $R \rightarrow$ reduces

8 water has weaker /weaker molecular forces



9) sum of clockwise moments = sum of anticlockwise moments

$$25x w = (3x50) + (2x20) ;$$

$$25 w = 190$$

$$W = 190/25$$

$$W = 7.6 \text{ N} ;$$

10) becomes more stable

11. radiation;

12. bulb should be large and made of thin material

13. $F = Ke / e ;$

$$= 12/100 \times 10/5$$

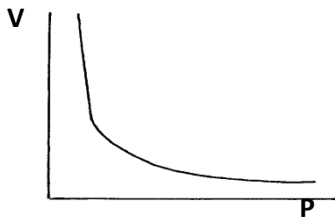
$$= 0.024 \text{ N/cm};$$

$$\text{Combined extension} = 5/3 + 5/2 + 5\text{N}$$

$$= 1.67 + 2.5 + 5 \text{ N}$$

$$= 9.15\text{N}$$

14.



15 a) a body remains at rest if moving or if moving continue to move in a straight line unless acted upon by an external force ;

b) i) horizontal = 20 mls but vertical = 0mls

$$S = \frac{1}{2} gt^2$$

$$20 = \frac{1}{2} \times 10 \times t^2$$



$$t^2 = 4$$

$$t = 2 \text{ sec;}$$

$$\text{ii) } R = ut$$

$$= 20 \text{ 82;}$$

$$= 40\text{m;}$$

$$\text{iii) } v = u + gt;$$

$$= 0+10-x2$$

$$= 20\text{mls}$$

$$\text{iv) } F = ma$$

$$= 50 \times 10 ;$$

$$= 500\text{N}$$

$$\text{v) } P = f_1/A_1 = F_2/A_2$$

$$500/50 = 25200/2520 \text{ at equilibrium}$$

The ratio is 10:10

The same = no change in position

$$16 \text{ a) i) } 40 \times 10^{-6} \text{ m}^3$$

$$\text{ii) } -277.5^0\text{e} + 2.5;$$

$$\text{iii) } u = 0.5 + 0.06$$

$$= 0.56\text{N;}$$

$$\text{B i) } U = 0.5 + 0.06 ;$$



$$= 0.56\text{N};$$

ii) weight of water displaced = weight of floating object

$$= 50 \times 10/1000$$

$$= 0.5\text{N};$$

Mass of water displaced = $0.5/10$

$$= 0.05 \text{ kg}$$

$$= V = m/\rho$$

$$= 0.05/1000;$$

$$= 5.0 \times 10^{-5} \text{ m}^3$$

iii) volume of object = $100 \times 58 \times 10^{-5}$

$$= 0.00025\text{m}^3$$

$$\rho = m/v$$

$$= 0.05/0.00025 ;$$

$$2000 \text{ kg/m}^3 ;$$

17 . i) At B the radius of the bend is small; more centripetal force is required ;

$$\text{ii) } F_r = WR$$

$$= 0.3 \times 60,000$$

$$= 18,000\text{N};$$

$$F_r = mv^2/r$$

$$= 18,000 = 60000v^2$$

$$V = \sqrt{18000 \times 25/60,000}$$

$$= 27.39 \text{ m/s};$$

$$\text{b) i) } T = 1/f ;$$

$$= 1/5$$

$$= 0.2 \text{ sec};$$

$$\text{ii) } W = 2\pi f ;$$



$$= 2 \times 3.142 \times 0.2;$$

$$= 1.257 \text{ rad s}^{-1}$$

$$c) V^2 = u^2 + 2as;$$

$$V^2 = 0.2 \times 0.28 \times h$$

$$V = \sqrt{0.56 \times 1.26}$$

$$= 0.84 \text{ m/s};$$

$$V = r\omega$$

$$0.84 = 0.14\omega;$$

$$\omega = 6 \text{ rad s}^{-1}$$

18 a) heat energy required to change a unit of a substance from liquid to vapour at a constant temperature

$$b) i) Q = mL$$

$$= \frac{3}{100} \times 2.226 \times 10^6$$

$$= 6780 \text{ J};$$

$$ii) Q = mc \Delta \theta$$

$$= \frac{3}{1000} \times 4200 (1000 - T)$$

$$= 1260 - 12.6T$$

$$iii) Q = MW C_W \Delta \theta + M_C C_C \Delta \theta$$

$$= 0.2 \times 4200 (T - 10) + 0.08 \times 900 (T - 10)$$

$$840 (T - 10) + 72 (T - 10)$$

$$840 T - 8400 + 72 T - 720$$

$$912 T - 9120$$

iv) heat gained = heat lost

$$912 T - 9120 = 1260 - 12.6T;$$

$$8924.6 T = 10380$$

$$T = 11.23^\circ \text{C};$$



19. a) $VR = 3$;

Efficiency $= M.A / VR \times 100$

$= 2.4$;

$M.A = \text{load} / \text{effort}$;

Effort $= 300 / 2.4 = 125\text{N}$

b) in one complete revolution the wheel moves at a distance $2\pi R$ and the axle (lead) moves a distance $2\pi r$;

$V.R = \text{Distance moved by effort} / \text{distance moved by load}$

$= 2\pi R / 2\pi r$;

$= R/r$;